

DSC 335, Handout 14: Logistic Regression (*Chapter 13*)

LEARNING OBJECTIVES.

- *Build a Bayesian regression model and simulate its posterior.*
- *Conduct inference using Bayesian regression models.*
- *Evaluate Bayesian regression models.*

Odds and Probability

Bayesian Logistic Regression Model

Example 1. We want to model $Y = \begin{cases} 1 & \text{if the student prefers 8AM to 3PM classes} \\ 0 & \text{otherwise} \end{cases}$ as a function of a student's college GPA using our class data set.

- (a) Construct and interpret a jitter plot of the data.
- (b) Fit the frequentist logistic regression model and give the equation of the regression line.
- (c) Write down the Bayesian regression model and determine a set of reasonable priors.
- (d) Fit this model in R and write down the regression equation based on posterior estimates.
- (e) Calculate and interpret 90% posterior credible intervals of both β_1 and e^{β_1} .
- (f) Plot 100 posterior plausible models for the probability that a student will prefer an 8AM class and describe what the graph says about the relationship between our variables.
- (g) Create a graph of the posterior predictive distribution of a student with a GPA of 3.8. Also give a point estimate for the probability that a student with a 3.8 GPA prefers 8AM classes.

Classification

Example 2. Using a classification cutoff of 0.5, what would we classify a student with a GPA of 3.8? What about a classification of 0.8?

Model evaluation for Logistic Regression

Example 3. Complete a posterior predictive check of our model and summarize the results.

Sensitivity and Specificity

Let Y be a set of binary response values on n data points and $\hat{Y}$ represent the corresponding posterior classifications of Y . A confusion matrix summarizes the results of these classifications relative to the actual observations where $a + b + c + d = n$:

The model's **overall accuracy** captures the proportion of all Y observations that are accurately classified:

Further, the model's **sensitivity** (true positive rate) captures the proportion of $Y = 1$ observations that are accurately classified and **specificity** (true negative rate) the proportion of $Y = 0$ observations that are accurately classified:

Example 4. Using a classification cut-off of 0.5, create and interpret a confusion matrix for our model.

Example 5. Now suppose we want to include whether or not a student is an athlete and the number of times a student works out a week, as well as an interaction between the two, as predictors in our model. Fit and evaluate this model using default priors. Compare the overall accuracy of the two models using four-fold cross-validation.